

Predicting Credit Default Risk

This assignment is group work.

NOTE: You must submit both **a Jupyter notebook** containing your code and **a separate written report**.

Project Overview

This project explores credit risk patterns in loan application data.

The dataset consists of two CSV files:

- `creditrisk_applications_final.csv`: the required dataset containing loan application, borrower, credit profile, and loan outcome information.
- `creditrisk_inquiries_final.csv`: an additional inquiry-level dataset containing supplementary credit inquiry information.

The datasets consist of **historical records** of completed loan application and repayment outcomes. Both datasets are expected to play a role in different parts of the analysis.

Each row in the main dataset represents a credit application with information about the requested loan, borrower characteristics, credit profile, and final loan outcome.

Your main objective is to build a machine learning **classification model** that predicts whether a **loan application results in default**.

Define the prediction target as:

- `target_default = 1` if `outcome_status == "Charged Off"`
- `target_default = 0` otherwise

You will need to create this target variable from the dataset before building your model.

General Guidelines for the Project

Your work should demonstrate a complete data analytics workflow and clearly communicate the results.

You may structure your project freely as long as these key components are included.

Data Cleaning

Prepare the dataset for meaningful analysis.

- Identify and handle missing values.
- Review and remove duplicates, inconsistencies, or invalid records.
- Ensure all columns use appropriate data types.

- Clean numeric fields stored as text.
- Standardize categorical values where needed.
- Document any assumptions or decisions you make.

In your report, clearly describe:

- The major cleaning steps you performed
- Any issues you discovered in the dataset
- How you handled missing or inconsistent values

Exploratory Data Analysis (EDA)

Develop a solid understanding of the dataset.

- Compute summary statistics for relevant features.
- Explore the distribution of the target variable.
- Examine how different borrower, loan, and credit attributes relate to default risk.
- Identify correlations, trends, or unusual observations.
- Investigate whether default risk appears to vary across loan purpose, risk grade, housing status, region, or other meaningful categories.

Your goal in this section is to develop intuition about the structure of the dataset and the factors that may influence default risk.

Not every variable explored during EDA must be used in the final model. However, this stage should guide your later decisions in feature selection and modeling.

In your report, present the most meaningful findings, not every calculation.

Visualization

Use visual tools to support and communicate insights.

Include plots such as bar charts, histograms, boxplots, correlation heatmaps.

Make sure visualizations:

- Have clear titles, proper names, and axis labels
- Are easy to interpret
- Help communicate something meaningful about credit risk

Your report should include figures that illustrate important patterns and explain what they show.

Feature Engineering

Prepare the dataset for modeling.

This may include:

- Creating new features from existing variables
- Encoding categorical variables
- Converting mixed text/numeric fields into usable numeric features
- Normalizing or scaling numeric features where appropriate
- Selecting features that are appropriate for prediction

You should think carefully about **which variables** are appropriate for prediction and explain your reasoning. Some variables in the dataset may be too closely tied to the prediction target. Some fields may be more useful for understanding the dataset than for building the final model.

In your report, **describe the main features you selected, any features you chose not to use, and the reasoning behind those decisions.**

Model Building

Develop classification models using the cleaned dataset.

You should build **at least two different models**.

Typical steps include:

- Splitting the data into training and testing sets
- Preparing features for modeling
- Training models on the training data
- Generating predictions on the test set

Carefully consider how the data should be divided for training and evaluation, and justify your approach.

In your report, briefly describe each model and the motivation for using it.

Model Evaluation

Assess model performance carefully.

Use appropriate evaluation metrics for binary classification.

You should:

- Compare the performance of different models
- Discuss strengths and weaknesses of each model
- Examine classification errors where relevant
- Interpret results in the context of credit default risk

It is normal for models to perform imperfectly on real-world data.

In your report, explain what the evaluation results mean and which model you would recommend. You should clearly interpret them.

Business Insights

Summarize your findings in a clear business context.

Your report should:

- Highlight the most important factors associated with default risk
- Explain which applicants or loan characteristics appear riskier
- Discuss how the model could support credit-risk decision-making
- Acknowledge limitations, assumptions, and data quality issues
- Suggest possible improvements or future analysis

Recommendations should be practical, evidence-based, and connected to the goal of understanding or predicting default risk.

Report Writing and Presentation

Write your report clearly, logically, and professionally.

Your report should include:

- Executive summary
- Description of the dataset and target variable
- Cleaning and preparation steps
- Key EDA findings
- Feature engineering decisions
- Model comparison and evaluation
- Business insights and recommendations
- Limitations and future work

Code Quality and Reproducibility

Ensure your code is professional, clear, and easy to follow.

- **Use a Jupyter notebook to write and run your code.**
- Properly comment your code and use markdown cells to explain key steps.
- Organize the notebook with clear section headings.
- Make sure your code can be run by someone else without modification.
- Avoid unnecessary duplication or unused code.

If your code requires anything special to run, briefly describe it in your report.

Appendix: Data Dictionary

creditrisk_applications_final.csv

Column Name	Description
application_id	Unique application identifier
requested_amount_usd	Requested loan amount in USD
repayment_term_text	Loan repayment duration as text
interest_rate_text	Interest rate associated with the loan
monthly_installment_usd	Scheduled monthly payment in USD
last_payment_amount_usd	Recorded last payment amount in USD
risk_grade_band	Broad risk grade category
risk_grade_detail	Detailed risk sub-grade

employment_tenure_text	Borrower employment length as text
housing_status	Borrower housing status
annual_income_usd	Reported annual income in USD
income_verification_status	Income verification category
origination_month	Loan origination month
outcome_status	Final loan outcome; used to create the prediction target
loan_purpose	Borrower-stated loan purpose
region_code	Regional or state code
debt_to_income_ratio	Borrower debt-to-income ratio
credit_score_low	Lower bound of credit score range
credit_score_high	Upper bound of credit score range
open_credit_lines	Number of open credit accounts
public_records_count	Number of derogatory public records
revolving_utilization_text	Revolving credit utilization value

creditrisk_inquiries_final.csv

Column Name	Description
requested_amount_usd	Requested amount in inquiry
application_date	Inquiry date
request_title	Free-text request title
screening_score	Screening or risk score field
debt_to_income_text	Debt-to-income field as text
region_code	Regional or state code
employment_tenure_text	Employment length as text
policy_code	Policy code indicator

Good luck!